IGCSE GRADE 9

Mathematics

Comprehensive Study Notes

Algebra • Geometry • Basic Calculus

Prepared by an Expert IGCSE Tutor with 20 Years of Experience

Cambridge IGCSE (0580) | Grade 9 Core & Extended

How to Use These Notes

These notes have been written specifically for IGCSE Grade 9 students following the Cambridge IGCSE Mathematics (0580) syllabus. Each chapter is structured to build understanding progressively — from core definitions through worked examples to exam-style practice.

■ **Blue Formula Box**	Key formulae you MUST memorise
✎ **Green Example Box**	Fully worked examples with step-by-step solutions
■ **Gold Tip Box**	Examiner tips and mark-winning strategies
■ **Red Warning Box**	Common mistakes that cost marks

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Chapter 1

Algebra

Algebra is the language of mathematics. It allows us to represent unknown quantities with letters, form equations, and solve real-world problems systematically. In IGCSE Grade 9 you are expected to handle expressions, equations, inequalities, indices, and sequences with confidence.

1.1 Algebraic Expressions & Simplification

An **algebraic expression** is a combination of numbers, variables, and operations. Simplifying means collecting like terms — terms that have identical variable parts.

Key Definitions:

Concept	Explanation
Term	A single number, variable, or product: $3x$, $-5y^2$, 7
Like Terms	Terms with the same variable(s) and power(s): $4x$ and $-2x$
Coefficient	The numerical factor of a term: in $7x^2$, the coefficient is 7
Expression	A group of terms joined by + or –: $3x^2 - 2x + 5$
Equation	Two expressions set equal: $3x + 2 = 11$

✎ **Simplify: $5x + 3y - 2x + 7y - 4$**

Step 1: Group like terms → $(5x - 2x) + (3y + 7y) - 4$

Step 2: Collect → $\mathbf{3x + 10y - 4}$

■ **Exam Tip**

Always check the sign BEFORE a term. The sign belongs to the term that follows it. Rewriting with brackets around each group helps avoid sign errors.

1.2 Expanding & Factorising

Expanding Brackets

Use the distributive law: $a(b + c) = ab + ac$

$$\mathbf{a(b + c) = ab + ac}$$

FOIL — Expanding Double Brackets

$(a + b)(c + d) = ac + ad + bc + bd$ — multiply each term in the first bracket by each term in the second.

$(a + b)(a - b) = a^2 - b^2$ (Difference of Two Squares)
$(a + b)^2 = a^2 + 2ab + b^2$

✎ **Expand $(2x + 3)(x - 5)$**

$= 2x \cdot x + 2x \cdot (-5) + 3 \cdot x + 3 \cdot (-5)$

$= 2x^2 - 10x + 3x - 15$

$= \mathbf{2x^2 - 7x - 15}$

Factorising

Factorising is the reverse of expanding. Always look for the **Highest Common Factor (HCF)** first.

✎ **Factorise $6x^2 - 9x$**

HCF of $6x^2$ and $9x$ is $3x$

$6x^2 - 9x = \mathbf{3x(2x - 3)}$

✎ **Factorise $x^2 - 16$**

Recognise: difference of two squares, $a = x$, $b = 4$

$x^2 - 16 = \mathbf{(x + 4)(x - 4)}$

■ **Common Mistake**

Do NOT expand and then try to refactorise in the same question — read what the question asks. If it says 'factorise', leave your answer in factored form.

1.3 Solving Linear Equations

A **linear equation** contains variables only to the power 1. The goal is to isolate the variable by performing the same operation on both sides.

Golden Rule: Whatever you do to one side, do to the other.

Step	Action
Step 1	Expand any brackets
Step 2	Collect variable terms on one side
Step 3	Collect constant terms on the other side
Step 4	Divide both sides by the coefficient of the variable

Step 5	Check by substituting back into the original equation

✎ **Solve $3(2x - 4) = 2x + 8$**

Expand: $6x - 12 = 2x + 8$

Rearrange: $6x - 2x = 8 + 12$

Simplify: $4x = 20$

Divide: $x = 5$

Check: $3(10-4) = 18$ and $2(5)+8 = 18$ ✓

1.4 Simultaneous Equations

Two equations with two unknowns. Methods: **Elimination** (add/subtract equations to remove one variable) or **Substitution** (express one variable in terms of the other).

✎ **Solve: $2x + 3y = 12$ and $x - y = 1$**

From equation 2: $x = y + 1$ (substitution)

Substitute: $2(y + 1) + 3y = 12 \rightarrow 2y + 2 + 3y = 12$

$5y = 10 \rightarrow y = 2$

$x = 2 + 1 = 3$

Solution: $\mathbf{x = 3, y = 2}$

■ **Exam Tip**

For the elimination method, multiply one or both equations to make coefficients of the same variable equal, then add or subtract. Always label your equations (1) and (2).

1.5 Quadratic Equations

A **quadratic equation** has the form $ax^2 + bx + c = 0$. Three solution methods:

Method	When to Use	Form
Factorising	Fastest when the quadratic factorises neatly	$(x+3)(x-2)=0 \rightarrow x=-3$ or $x=2$
Completing the Square	Useful for deriving the quadratic formula or finding vertex	$(x+p)^2 = q \rightarrow x = -p \pm \sqrt{q}$
Quadratic Formula	Always works; use when factorising is not obvious	$x = (-b \pm \sqrt{b^2-4ac}) / 2a$

$$x = (-b \pm \sqrt{b^2 - 4ac}) / 2a \quad \text{[Quadratic Formula]}$$

✎ **Solve $2x^2 + 5x - 3 = 0$ using the quadratic formula**

$a = 2$, $b = 5$, $c = -3$

Discriminant: $b^2 - 4ac = 25 - 4(2)(-3) = 25 + 24 = 49$

$x = (-5 \pm \sqrt{49}) / 4 = (-5 \pm 7) / 4$

$x = 2/4 = \mathbf{0.5}$ or $x = -12/4 = \mathbf{-3}$

■ **Common Mistake**

The discriminant ($b^2 - 4ac$) tells you about the roots: > 0 means two real roots; = 0 means one repeated root; < 0 means no real roots. Always calculate it first to know what to expect.

1.6 Inequalities

Inequalities are solved like equations with one crucial difference: multiplying or dividing by a **negative number reverses the inequality sign**.

If $-2x < 6$ then $x > -3$ (sign flips when dividing by -2)

Number Line Notation:

- ● Filled circle (•) for ≤ or ≥ (value is included)
- ■ Open circle (■) for < or > (value is excluded)

✎ **Solve $3 - 2x \leq 9$, $x \in$ ■, and list the integer solutions**

$-2x \leq 6$

$x \geq -3$ (inequality flips!)

Integer solutions: **–3, –2, –1, 0, 1, 2, …**

1.7 Indices & Surds

Laws of Indices:

Rule	Result	Example
a^m × a^n	a^(m+n)	x^3 × x■ = x■
a^m ÷ a^n	a^(m–n)	x■ ÷ x^2 = x■
(a^m)^n	a^(mn)	$(x^2)^3$ = x■
a^0	1	7■ = 1
a^(–n)	1/a^n	x^(–2) = 1/x^2
a^(1/n)	n-th root of a	8^(1/3) = 2

a^(m/n)	(n-th root of a)^m	27^(2/3) = 9

Surds:

A surd is an irrational root, e.g. √2, √5, √3. They cannot be simplified to an exact decimal. Key skills: simplify, rationalise the denominator.

√(ab) = √a × √b | √a / √b = √(a/b)

✎ **Rationalise the denominator: 5 / √3**

Multiply numerator and denominator by √3:

5/√3 × √3/√3 = 5√3 / 3 = **5√3/3**

1.8 Sequences & nth Term

A **sequence** is an ordered list of numbers following a rule. The **nth term** (T_n) is a formula giving the value of any term from its position.

Arithmetic Sequences (linear — constant difference d):

T_n = a + (n – 1)d where a = first term, d = common difference

Quadratic Sequences (second difference is constant):

Find the second differences; if constant = 2k, then the nth term starts with kn^2.

✎ **Find the nth term of: 5, 8, 11, 14, …**

d = 3, a = 5

T_n = 5 + (n–1)×3 = 5 + 3n – 3 = **3n + 2**

Check: n=1 → 5 ✓, n=4 → 14 ✓

■ **Exam Tip**

Always verify your nth-term formula by substituting n = 1, 2, and a large value. In the exam, list the first few terms from your formula to confirm.

Chapter 2

Geometry

Geometry explores the properties of shapes, lines, and angles. IGCSE Grade 9 requires both reasoning with angle facts and computing lengths and areas. Always give reasons for each geometric step in a proof.

2.1 Angles & Parallel Lines

Angle Fact	Property	Notation
Angles on a straight line	Sum = 180°	a + b = 180°
Angles around a point	Sum = 360°	a + b + c = 360°
Vertically opposite angles	Equal	a = b
Alternate angles (Z-angles)	Equal (parallel lines)	a = b
Co-interior angles (C-angles)	Sum = 180° (parallel lines)	a + b = 180°
Corresponding angles (F-angles)	Equal (parallel lines)	a = b

■ **Exam Tip**

In all geometry proofs, ALWAYS state your reason in words (e.g. 'alternate angles, AB ■ CD'). No reason = no mark, even if the calculation is correct.

2.2 Triangles & Congruence

The angle sum of a triangle is always 180°. An **exterior angle** of a triangle equals the sum of the two non-adjacent interior angles.

Exterior angle = sum of the two opposite interior angles

Congruence Conditions:

Condition	Meaning
SSS	Three sides equal
SAS	Two sides and the included angle equal
ASA / AAS	Two angles and a corresponding side equal
RHS	Right angle, hypotenuse, and one other side equal (right-angled triangles only)

Similarity:

Two shapes are similar if all corresponding angles are equal and sides are in the same ratio.

Scale factor k: lengths × k, areas × k^2, volumes × k^3

2.3 Polygons

Sum of interior angles of an n-gon = $(n - 2) \times 180°$

Each interior angle of a REGULAR n-gon = $(n - 2) \times 180° / n$

Each exterior angle of a REGULAR n-gon = $360° / n$

Polygon	Sides	Sum of Interior Angles	Each Angle (regular)
Triangle	3	180°	60°
Quadrilateral	4	360°	90°
Pentagon	5	540°	108°
Hexagon	6	720°	120°
Octagon	8	1080°	135°
Decagon	10	1440°	144°

2.4 Circles & Circle Theorems

Mastering circle theorems is essential for IGCSE. Each theorem has a specific reason that must be quoted in full.

Theorem	Statement
Angle at centre	Angle at centre = 2 × angle at circumference (same arc)
Angle in semicircle	Angle in a semicircle = 90°
Angles in same segment	Angles in the same segment are equal
Opposite angles, cyclic quad	Opposite angles of a cyclic quadrilateral sum to 180°
Tangent–radius	Tangent is perpendicular to radius at point of contact
Two tangents from a point	Tangents from an external point are equal in length
Alternate segment theorem	Angle between tangent and chord = inscribed angle in alternate segment

Chord bisector	Perpendicular from centre to a chord bisects the chord

■ **Exam Tip**

Draw a large, clear diagram for every circle theorem question. Label ALL known angles and use a different colour for the unknown. Always state the theorem name as your reason.

2.5 Perimeter, Area & Volume

Shape	Area / Surface Area	Volume / Circumference
Rectangle	$A = lw$	—
Triangle	$A = \frac{1}{2}bh$	—
Parallelogram	$A = bh$	—
Trapezium	$A = \frac{1}{2}(a+b)h$	—
Circle	$A = \pi r^2$	$C = 2\pi r$
Sector (angle θ°)	$A = (\theta/360)\pi r^2$	Arc $= (\theta/360)\times 2\pi r$
Cuboid	$SA = 2(lw+lh+wh)$	$V = lwh$
Cylinder	$SA = 2\pi rh + 2\pi r^2$	$V = \pi r^2 h$
Cone	$SA = \pi rl + \pi r^2$	V = ■$\pi r^2 h$
Sphere	$SA = 4\pi r^2$	$V = (4/3)\pi r^3$
Pyramid	—	V = ■ × base area × h

■ **Common Mistake**

Always check units and convert before calculating. Mixing cm and m in the same formula is one of the most common errors. State units in your final answer.

2.6 Trigonometry (SOH–CAH–TOA)

Trigonometry connects the angles and sides of right-angled triangles. Label sides relative to the angle θ: Opposite (O), Adjacent (A), Hypotenuse (H).

$\sin\theta = O/H$ $\cos\theta = A/H$ $\tan\theta = O/A$
SOH – CAH – TOA

Exact Values (Memorise!):

Angle θ	sin θ	cos θ	tan θ
0°	0	1	0
30°	1/2	$\sqrt{3}/2$	$1/\sqrt{3}$
45°	$1/\sqrt{2}$	$1/\sqrt{2}$	1
60°	$\sqrt{3}/2$	1/2	$\sqrt{3}$
90°	1	0	Undefined

✎ **Find the length x in a right triangle where the hypotenuse = 10 cm and θ = 35°**

We want the side Opposite to 35°, and we know H = 10

$\sin 35° = x / 10$

$x = 10 \times \sin 35° = 10 \times 0.5736 \approx$ **5.74 cm**

For non-right triangles — Sine & Cosine Rules:

Sine Rule: $a/\sin A = b/\sin B = c/\sin C$
Cosine Rule: $a^2 = b^2 + c^2 - 2bc \cdot \cos A$
Area of triangle: Area = $\frac{1}{2}ab \sin C$

2.7 Pythagoras' Theorem

$a^2 + b^2 = c^2$ where c is the hypotenuse

✎ **Find the hypotenuse when legs are 5 cm and 12 cm**

$c^2 = 5^2 + 12^2 = 25 + 144 = 169$

$c = \sqrt{169} =$ **13 cm**

Pythagorean Triples to Memorise:

3, 4, 5	**5, 12, 13**	**8, 15, 17**	**7, 24, 25**

2.8 Vectors

A **vector** has both magnitude (size) and direction. Written as a column vector or bold letter. The magnitude of vector (a, b) is $\sqrt{(a^2 + b^2)}$.

$$|v| = \sqrt{(a^2 + b^2)} \text{ for } v = (a, b)$$

Key Vector Operations:

Operation	Rule	Note
Addition	(a,b) + (c,d) = (a+c, b+d)	Add components separately
Scalar multiplication	k(a,b) = (ka, kb)	Multiply each component by k
Subtraction	(a,b) – (c,d) = (a–c, b–d)	Subtract components
Negative vector	–v reverses direction	Same magnitude, opposite direction

Chapter 3

Basic Calculus

Calculus is the mathematics of change and accumulation. At Grade 9 IGCSE level you are introduced to the key ideas of differentiation and integration — the two fundamental operations of calculus. These topics form the foundation for A-Level Mathematics and beyond.

3.1 Introduction to Functions

A **function** f maps each input value x to exactly one output f(x). We write f : x ■ f(x). The **domain** is the set of valid inputs; the **range** is the set of outputs.

$f(x) = 3x^2 - 2x + 1$ means 'the function of x is $3x^2 - 2x + 1$'

✎ If $f(x) = 2x^2 - 3$, find f(4) and f(–1)

$f(4) = 2(16) - 3 = 32 - 3 = \mathbf{29}$

$f(-1) = 2(1) - 3 = 2 - 3 = \mathbf{-1}$

Composite Functions:

fg(x) means apply g first, then f. Read right to left: f(g(x)).

fg(x) = f(g(x)) gf(x) = g(f(x))

Inverse Functions:

f■¹(x) reverses the operation of f. To find it: write y = f(x), swap x and y, then rearrange for y.

3.2 Limits — Intuitive Approach

The **limit** of a function f(x) as x approaches a value c is the value f(x) gets closer and closer to, without necessarily reaching it.

lim[x→c] f(x) = L means f(x) → L as x → c

✎ Find lim[x→3] $(x^2 - 9) / (x - 3)$

Direct substitution gives 0/0 — try factorising:

$(x^2-9)/(x-3) = (x+3)(x-3)/(x-3) = x + 3$ (for $x \neq 3$)

lim[x→3] = 3 + 3 = **6**

■ Exam Tip

Always try direct substitution first. If you get 0/0, factorise or simplify. The limit exists even if the function is undefined at that exact point.

3.3 The Gradient & Rate of Change

The **gradient of a curve** at a point is the slope of the tangent to the curve at that point. It represents the **instantaneous rate of change**.

Average gradient = (y■ – y■) / (x■ – x■) = Δy / Δx

As $\Delta x \to 0$, the average gradient approaches the **derivative**, written dy/dx or $f'(x)$.

3.4 Differentiation from First Principles

This is the formal definition of the derivative — how differentiation was originally defined.

$$f'(x) = \lim_{h\to 0} [\, f(x+h) - f(x) \,] / h$$

✎ Differentiate $f(x) = x^2$ from first principles

$f(x+h) = (x+h)^2 = x^2 + 2xh + h^2$

$f(x+h) - f(x) = 2xh + h^2$

$[f(x+h) - f(x)] / h = 2x + h$

As $h \to 0$: $f'(x) =$ **$2x$**

■ Exam Tip

In first principles questions, always show the limit notation and the cancellation of h. Full working is required for all the marks.

3.5 Differentiation Rules

Once the first principles are understood, use the standard rules for speed and efficiency.

Power Rule (most important):

d/dx [x^n] = n·x^(n–1) (Multiply by the power, reduce power by 1)

Rule	Formula	Example
Constant	d/dx [c] = 0	d/dx [7] = 0
Power rule	d/dx [x^n] = nx^(n-1)	d/dx [x■] = 5x■

Constant multiple	d/dx [cf(x)] = c·f'(x)	d/dx $[3x^2] = 6x$
Sum rule	d/dx [f+g] = f' + g'	d/dx $[x^2+x] = 2x+1$
Difference rule	d/dx [f–g] = f' – g'	d/dx $[x^3–4x] = 3x^2–4$

✎ **Differentiate $y = 4x^3 – 6x^2 + 5x – 2$**

$dy/dx = 4·3x^2 – 6·2x + 5·1 – 0$

$dy/dx =$ **$12x^2 – 12x + 5$**

✎ **Differentiate $y = 3/x^2 + 2\sqrt{x}$ (rewrite first!)**

$y = 3x^{(–2)} + 2x^{(1/2)}$

$dy/dx = 3·(–2)x^{(–3)} + 2·(½)x^{(–1/2)}$

$dy/dx =$ **$–6x^{(–3)} + x^{(–1/2)}$** or $–6/x^3 + 1/\sqrt{x}$

3.6 Applications of Differentiation

Tangents and Normals

The gradient of the tangent to y = f(x) at x = a is f'(a). The normal is perpendicular to the tangent: m_normal = –1 / m_tangent.

Tangent at (a, b): y – b = f'(a)(x – a)

Normal at (a, b): y – b = [–1/f'(a)](x – a)

✎ **Find the equation of the tangent to $y = x^2 – 3x$ at x = 4**

dy/dx = 2x – 3; at x = 4: gradient = 2(4)–3 = 5

y at x=4: y = 16 – 12 = 4 → point (4, 4)

Tangent: y – 4 = 5(x – 4) → **y = 5x – 16**

Stationary Points (Turning Points)

Stationary points occur where dy/dx = 0. Use the second derivative test to classify them.

At a stationary point: dy/dx = 0

Second Derivative	Type of Point	Shape
$d^2y/dx^2 > 0$	Minimum point	Curve is concave up (U-shape)
$d^2y/dx^2 < 0$	Maximum point	Curve is concave down (∩-shape)

$d^2y/dx^2 = 0$	Possible inflection	Check gradient either side

✎ Find and classify stationary points of $y = x^3 - 3x^2 - 9x + 5$

$dy/dx = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x-3)(x+1)$

Stationary points: $x = 3$ or $x = -1$

$d^2y/dx^2 = 6x - 6$

At $x = 3$: $d^2y/dx^2 = 12 > 0 \rightarrow$ **minimum**

At $x = -1$: $d^2y/dx^2 = -12 < 0 \rightarrow$ **maximum**

Kinematics — Motion in a Straight Line

If displacement s is given as a function of time t, calculus gives us velocity and acceleration directly.

$v = ds/dt$ (velocity = rate of change of displacement)

$a = dv/dt = d^2s/dt^2$ (acceleration = rate of change of velocity)

Object is at rest when $v = 0$. Object changes direction when $v = 0$ and changes sign.

3.7 Introduction to Integration

Integration is the reverse of differentiation — it is called the **anti-derivative**. The indefinite integral includes a constant of integration C (because differentiating any constant gives zero).

$\int x^n\,dx = x^{n+1} / (n+1) + C \quad (n \neq -1)$

Rule	Formula	Example
Constant	$\int a\,dx = ax + C$	$\int 5\,dx = 5x + C$
Power rule	$\int x^n\,dx = x^{n+1}/(n+1) + C$	$\int x^3\,dx = x$■$/4 + C$
Constant multiple	$\int cf(x)\,dx = c\int f(x)dx$	$\int 3x^2\,dx = x^3 + C$
Sum rule	$\int[f+g]dx = \int f\,dx + \int g\,dx$	$\int(x^2+x)dx = x^3/3+x^2/2+C$

✎ Find $\int(6x^2 - 4x + 3)\,dx$

$= 6 \cdot x^3/3 - 4 \cdot x^2/2 + 3x + C$

$=$ **$2x^3 - 2x^2 + 3x + C$**

■ **Common Mistake**

Never forget the constant of integration C in an indefinite integral. Its absence will cost you a mark. C is only determined when you have a specific boundary condition, such as 'the curve passes through (2, 5)'.

3.8 Area Under a Curve — Definite Integrals

The **definite integral** calculates the exact area between a curve, the x-axis, and vertical lines $x = a$ and $x = b$.

$$\int_a^b f(x)\,dx = [F(x)]_a^b = F(b) - F(a)$$

where $F(x)$ is the anti-derivative of $f(x)$.

✎ Find the area under $y = 3x^2$ between $x = 1$ and $x = 4$

$\int_1^4 3x^2\,dx = [x^3]_1^4$

$= 4^3 - 1^3$

$= 64 - 1$

$=$ **63 square units**

Important Notes on Area:

- If the curve is below the x-axis, the integral gives a negative value. Take the absolute value $|\int f(x)dx|$ for the actual area.
- If the area straddles the x-axis (partly above, partly below), split into two separate integrals at the x-intercept.

■ **Exam Tip**

Always sketch the curve before computing area. Identify where it crosses the x-axis and note whether regions are above or below. A diagram prevents sign errors.

Quick-Reference Formula Sheet

ALGEBRA

- Quadratic Formula: $x = (-b \pm \sqrt{(b^2-4ac)}) / 2a$
- Difference of Squares: $a^2 - b^2 = (a+b)(a-b)$
- Perfect Square: $(a\pm b)^2 = a^2 \pm 2ab + b^2$
- nth term (arithmetic): T_n = a + (n–1)d
- Laws of Indices: a^m × a^n = a^(m+n); a^(–n) = 1/a■; a^(1/n) = ■√a

GEOMETRY

- Polygon interior angle sum: $(n-2) \times 180°$
- Circle area: πr^2 | Circumference: $2\pi r$
- Cylinder volume: $\pi r^2 h$ | Cone volume: ■$\pi r^2 h$ | Sphere volume: $(4/3)\pi r^3$
- Pythagoras: $a^2 + b^2 = c^2$
- SOH–CAH–TOA | Sine rule: a/sinA = b/sinB | Cosine rule: $a^2 = b^2+c^2-2bc\cdot\cos A$
- Vector magnitude: $|v| = \sqrt{(a^2+b^2)}$ for v = (a,b)

CALCULUS

- Derivative (power rule): d/dx[x■] = nx■■^{1}
- Integral (power rule): ∫x■ dx = x■■^{1}/(n+1) + C
- Tangent at (a,b): y – b = f'(a)(x – a)
- Stationary points: dy/dx = 0
- Definite integral: ∫[a→b] f(x)dx = F(b) – F(a)
- Kinematics: v = ds/dt, a = dv/dt

Final Exam Revision Strategy

4 Weeks Before

✓ Complete one full past paper under timed conditions each week

✓ Identify your 3 weakest topics — dedicate focused sessions to each

✓ Practise all circle theorems: write each theorem and draw a diagram from memory

✓ Drill exact trig values until they are instant recall

2 Weeks Before

✓ Rework every question you got wrong in past papers — understand WHY you erred

✓ Make flashcards for every formula; test yourself daily

✓ Practice showing all working — IGCSE awards method marks

✓ Focus on multi-step problems: simultaneous equations, quadratics, calculus applications

Final Week

✓ Do one full paper each day under strict exam conditions

✓ Review your formula sheet — make sure every item is memorised

✓ Read questions twice before answering. Highlight key words: 'exact', 'hence', 'show that'

✓ Get 8 hours of sleep. Your brain consolidates memory during sleep

These notes cover Cambridge IGCSE Mathematics (0580) Grade 9 content. Always cross-reference with your official Cambridge syllabus document. Past papers are available at cambridgeinternational.org. Good luck — you've got this!